6) a) If the sum of Right hand poles (+), clockwise encirclements of (-1,0) (+), and counter clockwise encirclements of (-1,0) (-) is >= 0 then the system is unstable. There are 2 RHP poles, and 1 N_{CCW}, ~~[illegible]~~ adding up to 1. So the closed loop system will be unstable.

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b) $G_{OL}(s) = \frac{s^2 - 18s - 10}{s^2 - 5s + 4}$

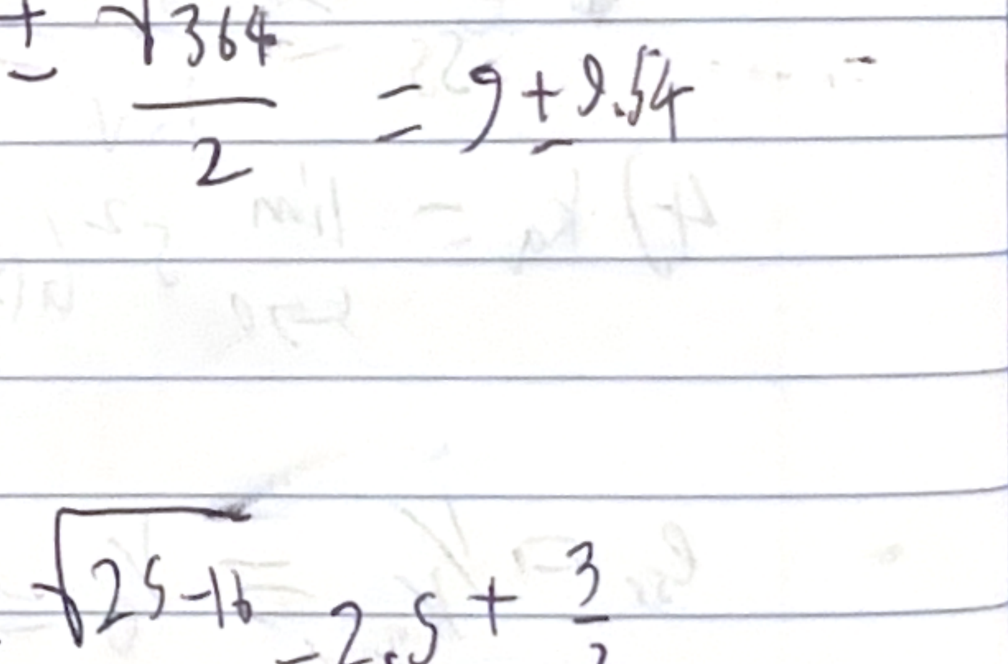
1) Zeros: $\frac{+18 \pm \sqrt{18^2 - 4(1)(-10)}}{2(1)} = 9 \pm \frac{\sqrt{364}}{2} = 9 \pm 9.54$

Zeros = 18.54, -0.54

2) $\frac{5 \pm \sqrt{(-5)^2 - 4(1)(4)}}{2(1)} = 2.5 \pm \frac{\sqrt{25-16}}{2} = 2.5 \pm \frac{3}{2}$

poles: 4, 1

3)



4) Both poles are positive ∴ The open loop system is unstable

c) $G_{CL}(s) = \dfrac{\dfrac{s^2-18s-10}{s^2-5s+4}}{1+\dfrac{s^2-18s-10}{s^2-5s+4}(1)} \cdot \dfrac{s^2-5s+4}{s^2-5s+4} = \dfrac{s^2-18s-10}{(s^2-5s+4)+(s^2-18s-10)}$

$= \dfrac{s^2-18s-10}{2s^2-23s-6}$

1) zeros are the same: **zero = 18.54, −0.54**

2) poles: $\dfrac{-23 \pm \sqrt{(-23)^2-4(2)(-6)}}{2(2)} = \dfrac{-23}{4} \pm \dfrac{\sqrt{577}}{4} = -5.75 \pm 6$

poles = 0.25, −11.75

3)

4 B) There is still a right hand pole ∴ the system is still unstable

